

Julian Cabrera, Ph.D.

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Professional Summary

Seismic Imaging Expert & Technical Leader | 38+ Years Experience

With over three decades of expertise in seismic imaging and processing, I bring a unique combination of technical leadership and hands-on problem-solving capabilities to exploration and production challenges. My career encompasses:

- **Technical Supervision Excellence:** Led complex 3D seismic imaging projects across deep/shallow-water marine and land environments, navigating challenging geological settings including salt and mobile shale tectonics. Effectively managed deliverables from major contractors including CGG, SLB, ION, and PGS, consistently delivering exploration-ready products aligned with geological expectations.
- **Advanced Model Building Oversight:** Ensured the correct application of cutting-edge technologies including Full Waveform Inversion (FWI) through collaborative workflows with E&P geologists and geophysicists, bridging the gap between technical contractors and exploration teams to achieve successful imaging objectives.
- **Critical Analysis Capabilities:** Specialized in analytical evaluation and troubleshooting of seismic images and velocity models for prospects and drilling locations, directly contributing to exploration success and risk reduction in complex geological settings.
- **Knowledge Transfer Leadership:** Established successful training programs for geologists and geophysicists in seismic imaging technologies and workflows, elevating overall team capabilities.
- **High-Performance Computing Innovation:** Developed production-grade, high-performance seismic processing algorithms with particular emphasis on seismic imaging applications, leveraging scientific programming expertise to solve practical exploration challenges.

My comprehensive geophysical background spanning technical project supervision and leadership, direct collaboration with exploration teams, and algorithm development enables me to deliver exceptional value to E&P companies through both technical excellence and strategic problem-solving.

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Professional Experience

Geophysical Consultant

Seistech Imaging Inc. (Founder and Owner) | 2003-Present

Lead technical consultant specializing in seismic imaging and velocity model building for complex geological environments. Supervise projects, identify shortcomings, propose remedies, and ensure the correct implementation of imaging workflows that are consistent with the geological model to enhance exploration outcomes. Bridge the gap between technical contractors and exploration teams to ensure delivery of geologically meaningful results.

Professional Philosophy: My approach centres on ensuring technical quality and geological plausibility in seismic imaging products, applying thorough quality control processes while maintaining flexibility to address unique exploration challenges. I focus on translating technical complexities into practical decision-making insights for exploration teams.

Client Engagements:

Pemex E&P, México | 2003-2019 & 2022-2024

- Provide expert technical supervision of well over 30,000 km² 3D seismic imaging projects in complex salt and mobile-shale tectonic environments, targeting Cretaceous-Jurassic and Lower Tertiary exploration objectives. Projects span deep and shallow water, and land areas (e.g., over the Perdido Fold Belt and Sureste Basin). Marine data are from FAZ, WAZ, NAZ and OBS acquisitions.
- Designed and implemented, for a Pemex Asset team, interpretation-based de-migration/re-migration and seismic image-enhancement processing workflows that increased interpretation confidence of prospects and exploration opportunities in the Sureste Basin.
- Conduct quality assurance on behalf of Pemex of deliverables from its seismic contractors, including CGG, Schlumberger, ION, PGS, Fugro and Paradigm among others, ensuring technical specifications are met while assuring geological consistency of the velocity models and seismic images.
- Prepared and teach training courses on Seismic Depth Imaging and on Full-waveform Inversion specifically designed for geologists and geophysicists of Exploration teams.
- Authored Pemex's standard seismic imaging sequence used in its bidding process for processing projects, emphasizing integrated geophysical-geological workflows that became the company standard.

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Schlumberger, Mexico & Houston | 2017-2024

- Serve as an in-house Seismic Imaging Consultant for Pemex and Multi-Client projects in the Gulf of Mexico, reporting directly to the President and the VP of the North America-Mexico Geoscience division, and to the Geophysical manager of its Mexican section.
- Perform analysis of seismic images and velocity models, identifying issues, recommending targeted technical solutions, and elevating the geological accuracy of seismic imaging products supporting oil exploration and lease round sales.
- Facilitated effective communication between processing teams and client exploration departments, ensuring alignment on technical approaches and exploration objectives.

Compañía Mexicana de Exploraciones, México | 2009-2010

- Directed technical quality assurance for seismic imaging and converted-wave processing projects, establishing QC protocols that improved delivery consistency.

Fugro Seismic Imaging, Houston | 2009-2010

- Led velocity model building and depth migration projects in complex geological settings.
- Provided expert troubleshooting for seismic imaging and model building challenges, reducing project delays and improving final product quality.

TGS Imaging, Houston | 2007

- Developed and implemented a high-performance 3D SRME multiple-adaptive subtraction algorithm for production-grade seismic processing.

Seismic Imaging Consultant

PDVSA E&P, Caracas, Venezuela | 1998-2002

- Managed quality assurance for seismic imaging projects executed by PDVSA's contractors, including CGG and SLB.
- Ensured technical specifications and imaging objectives were consistently met and in line with the Exploration teams requirements, and throughout the project life-cycle supporting PDVSA exploration success.

Seismic Imaging Research Consultant

PDVSA Intevep (Research Division), Los Teques, Venezuela | 1996-1998

- Developed high-performance seismic imaging algorithms for the IBM-SP2 supercomputer, translating research concepts into practical exploration tools.
- Established and delivered training programs for research geophysicists in High Performance Computing (HPC) software development for seismic applications.

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Sr. Research Geophysicist

Western Geophysical, Houston | 1993-1996

- Designed and implemented advanced seismic imaging algorithms optimized for parallel-processing on SGI and Connection Machine HPC platforms.
- Created computational solutions for processing large 3D seismic datasets in a cost-effective and production-level environment, contributing to Western's technological leadership of leading-edge imaging algorithms.

Sr. Research Geophysicist

Tensor Geophysical, New Orleans | 1990-1993

- Pioneered development of the industry's first commercial 3D Kirchhoff Prestack depth migration algorithm for the Intel Hypercube supercomputer.
- Implemented complete algorithmic solution from mathematical and geophysical concepts to production-ready code. Closely interacted with Amoco Research's senior staff of Tulsa, OK, who provided quality assurance of my code implementation.

Sr. Research Geophysicist

Inverse Theory and Applications, Vancouver & Calgary | 1984-1990

- Co-founded innovative startup offering first-of-its-kind interactive seismic processing workstation.
- Developed comprehensive suite of geophysical algorithms including tau-p transforms, seismic migrations, and inversion techniques.

Geophysicist

Mexican Petroleum Institute, México City | 1981

- Performed seismic and gravity data processing for exploration projects.

Geophysicist

Geophysical Service Inc., Mexico City | 1978

- Conducted seismic data processing.

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Education

- **PhD, Geophysics | University of British Columbia, Vancouver, Canada (1990)**
 - Dissertation: "Velocity-dip analysis and prestack imaging in the plane-wave domain."
- **M.Sc., Geophysics | University of British Columbia, Vancouver, Canada (1984)**
 - Thesis: "Plane-wave decomposition and spherical-wave reconstruction as a linear inverse problem."
- **Engineering, Geophysics | National Polytechnic Institute, Mexico City, Mexico (1981)**
 - Thesis: "Directional filtering of gravity and magnetic data."

Skills

Geophysical Expertise

- **Seismic Imaging & Processing:** Advanced velocity model building, FWI and Tomography techniques, industry-standard migration algorithms.
- **Quality Control & Analysis:** Systematic evaluation of seismic imaging workflows and algorithms; assessment of geological consistency of velocity models and seismic images.

Project Management & Technical Supervision

- **Seismic Contractor Management:** Technical oversight of major service providers executing seismic imaging workflows for O&C clients.
- **Cross-functional Collaboration:** Bridging seismic processing and exploration teams, facilitating geophysical-geological integration.

High-Performance Computing

- **Programming Languages:** C/C++, Fortran (scientific computing), Python.
- **Parallel Computing Frameworks:** OpenMP, CUDA for GPUs.
- **Algorithm Development:** Seismic imaging and processing, Machine learning.

Visualization & Processing Tools

- OpendTect, Focus & Geodepth (Echos).

Professional Recognition

- **SEG Honorary Lecturer, Central and South America Region (2011)**
 - Lecture: "The State and Future Directions of Prestack Depth Migration."

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Continuing Education

- Certified C++ Associate Programmer, C++ Institute (2020).
- Machine Learning, Deep Neural Networks and Convolutional Neural Networks, Coursera Certifications (2019).
- Advanced Sequence Stratigraphic - Applications for Exploration, AAPG (2016).
- Deep-Water Salt Tectonics, AAPG (2008).
- Java Technology Fast Track Program, Sun Microsystems (2000).

Professional Affiliations

- Society of Exploration Geophysicists.

Additional Information

- Citizenship: Canada, United States and México.
- Fully bilingual: English and Spanish.

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Papers Presented at International Conferences

- Cabrera, J., 2015, Fundamentals of the Depth Migration PSDM process for interpreters: Colombian Congress of Geology, SEG invited speaker. Bucaramanga, Colombia.
- Ponce, R., Vazquez, A., Cabrera, J., Salazar, H., Ronzon, I., and Cabrales, A., 2012, Current state of depth imaging in Mexico: Mexican Petroleum Congress, Mexico City.
- Cabrera, J., Vera, J., and Yepes, S., 2003, Application of 3D Prestack Time Migration to improve the seismic image of Cretaceous target structures found in the tectonic belt of the *Cosomapa* area, Veracruz State: 73th. Int'l Mtg. of the Society of Exploration Geophysicists, Denver.
- Clemente, G., Cabrera, J., Carrillo, I., and Perez Cruz, G., 2003, Best practices for the construction of the seismic migration velocity field in an ultra-deep water Gulf of Mexico project: 1st International Geosciences Congress of the Petroleum Industry, Cancun, Mexico.
- Goncalves, A., Cabrera, J., and Carneiro, V., 2002, Poststack depth migration in the *Furrial* Oil Field: 11th Venezuelan Geophysical Congress, Caracas, Venezuela.
- Grandi, S., and Cabrera, J., 2000, Prestack time migration applied in the *Piedritas* area, Eastern Venezuela - A Case Study: 10th Venezuelan Geophysical Congress, Caracas, Venezuela.
- Cabrera, J., Celis, V., Amaricua, S., and Araujo, M., 1998, Experiences in depth imaging of Venezuela data: 9th Venezuelan Geophysical Congress, Caracas, Venezuela.
- Cabrera, J., 1996, Tutorial on depth imaging: 8th Venezuelan Geophysical Congress, Maracaibo, Venezuela.
- Cabrera, J., and Wiggins, W., 1996, Benefits and limitations of 3-D poststack depth imaging in complex geological settings: II Int'l Congress & Exhibition of the American Association of Petroleum Geologists/ Sociedad Venezolana de Geólogos, Caracas, Venezuela.

Publications

- Cabrera, J., Celis, V., Amaricua, S., and Araujo, M., 1998, Experiences in depth imaging of Venezuela data: The Leading Edge, Special issue on Venezuela.
- Cabrera, J., and Levy, S., 1989, Shot DMO with logarithmic transformations: Geophysics, v.54, 1038-1041.
- Chapman, R., Stinson, K., Levy, S., Cabrera, J., and Oldenburg, D., 1988, Estimation of the elastic properties of sea floor sediments by inversion of pre-critical reflection data: IEEE Journal of Oceanic Engineering, v.13, 215-221.
- Levy, S., Cabrera, J., Stinson, K., Oldenburg, D., and Chapman, R., 1987, The estimation of density, P-wave and S-wave speeds of the top-most layer of sediments: Progress in Underwater Acoustics, Plenum Publishing Co.
- Cabrera, J., and Levy, S., 1984, Stable plane-wave decomposition and spherical-wave reconstruction-applications to converted S-wave separation and trace interpolation: Geophysics, v.49, 1915-1932.

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Cabrera, J., Perkins, W., Ratcliff, D., and Lynn, W., 1992, 3-D prestack depth migration on a massively parallel computer- implementation and case history: 62nd Int'l Mtg. of the Society of Exploration Geophysicists, New Orleans.

Cabrera, J., and Levy, S., 1989, Velocity-dip analysis in the plane-wave domain: 51st Int'l Mtg. of the European Association of Exploration Geophysicists, Berlin.

Cabrera, J., and Levy, S., 1988, Shot DMO with logarithmic transformations: 4th Venezuelan Geophysical Congress, Caracas, Venezuela.

Cabrera, J., Levy, S., and Stinson, K., 1985, A fast algorithm for plane-wave decomposition: 55th Int'l Mtg. of the Society of Exploration Geophysicists, Washington, DC.